

WiFi 6 Promises Much More than Faster Speeds

WiFi 6 increases the network connectivity, and has been developed to ensure the trillions of devices connected in the near future continue to work seamlessly and efficiently. Though it was certified in 2019, it faced a few challenges in testing due to the pandemic. This article gives an overview of this technology.



The next-generation standard in WiFi technology, termed 'WiFi 6', is also known as 'AX WiFi' or '802.11ax'. Developed to meet the exponential increase in demand for devices in the world, it can be used for virtual reality (VR) and smart home devices as well. It is an improvement on the current 802.11ac WiFi standard and meets current challenges in terms of capacity, efficiency, coverage and performance.

Researched in 2014, this standard was invented in 2018 and launched by the IEEE High Efficiency WLAN Study Group (HEW SG). It began certifications in late 2019, with Samsung's Galaxy Note 10 and Ruckus R750 employing this technology. Specified to operate between the 1GHz and 6GHz bands, WiFi 6 focuses mainly between the 2.4GHz and 5GHz frequencies.

In an average household today,

nine devices are connected to WiFi. WiFi 6 aims at improving the network rather than boosting the speed for individual devices.

Features of WiFi 6

Multi-user, multi-input, multi-output (MU-MIMO): This communication between routers and multiple devices concurrently. It supports four simultaneous data streams, added to which one user can have a considerable bandwidth of incoming data from a smart router, both on the 2.4GHz and 5GHz frequencies.



Figure 1: WiFi 6

1024-QAM: This helps WiFi 6 encode more bits per packet. There is a 25 per cent increase in throughput. Not only does it improve efficiency in high-traffic situations, it also maximises data rates. This is a huge advantage for modern enterprise applications.

Orthogonal frequency-division multiplexing (OFDM): This allows four times as many subcarriers and increases speed by 11 per cent. The expanded signal allows for greater simultaneous packet delivery across users. Hence, the wait time between packets and latency is reduced.

Increased channel width: The 160MHz channel communication is added to the 80MHz band, thus doubling the channel width. This allows routers to handle more users and provide larger streams per user.